

TOWARD LOW-ALTITUDE AIRSPACE MANAGEMENT AND UAV OPERATIONS: REQUIREMENTS, ARCHITECTURE AND ENABLING TECHNOLOGIES

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ABSTRACT

The low-altitude economy (LAE) is rapidly advancing toward intelligence, connectivity, and coordination, bringing new challenges in dynamic airspace management, unmanned aerial vehicle (UAV) operation, and security management. Existing systems remain fragmented and lack effective coordination. To bridge these gaps, we propose UTICN (Ubiquitous and Trusted Intelligent Cellular-native Network) for LAE, a unified cellular-native architecture that integrates multi-domain sensing, high-precision positioning, intelligent aircraft-to-everything communication, dynamic airspace management, and UAV operational services. UTICN introduces key technologies such as integrated sensing and communication (ISAC), passive and active positioning, intelligent machine communication, swarm coordination, and control-data decoupled management frameworks. We demonstrate UTICN's feasibility through two use cases, i.e., a city-level LAE management platform and a multi-frequency collaborative ISAC system. This work provides a fundamental reference for building a unified operational foundation and airspace management architecture for the LAE.

I. INTRODUCTION

The low-altitude economy (LAE) is emerging as a pivotal strategic domain for the future utilization of airspace resources [1]. A cornerstone of this advancement is the development of LAE networks (LAENets), which should be a ubiquitous, intelligent, and interoperable network infrastructure capable of seamlessly connecting unmanned aerial vehicles (UAVs) with terrestrial digital ecosystems [2], as well as providing airspace and security management. LAENets are crucial in supporting a diverse range of mission-critical applications, including urban logistics, infrastructure monitoring, aerial surveillance, electrically powered vertical takeoff and landing aircraft, and emergency

rescue [3]. In addition, recent industry reports project rapid economic and technological growth in this sector. For instance, the Civil Aviation Administration of China (CAAC) forecasts that the domestic low-altitude economy will exceed 1 trillion RMB by 2035, supported by comprehensive airspace reform policies. Similarly, GSMA Intelligence emphasizes that urban air mobility and UAV-based services will be a key driver for 5G-Advanced and 6G network demand, reinforcing the necessity of interoperable and trusted LAE infrastructures.

These applications impose stringent requirements on dynamic airspace management, UAV operational service, and safety assurance [4]. To meet these demands, LAENets must offer persistent spatiotemporal environmental sensing, high-precision positioning, mission-critical communication, and scalable cross-domain management and service capabilities [5]. However, current architectures fall short due to fragmented design, limited sensing capabilities, and insufficient cross-system coordination. For example, small UAVs are difficult to detect in dense urban environments; coordination across flight zones or jurisdictions remains inefficient; and existing systems lack timely response mechanisms for anomalies or security events.

To meet these growing demands, the LAE is evolving along three key technological trajectories. First, intelligence enables UAVs to perceive and make decisions through onboard artificial intelligence (AI). Second, connectivity is enabled through a combination of wide-area air-ground integration and heterogeneous technologies, including 5G-advanced (5G-A), reduced capability (RedCap), and aircraft-to-everything (A2X) communication. Third, effective system-level collaboration necessitates real-time coordination between UAV swarms, airspace infrastructures, and key stakeholders, including government authorities, UAV pilots, and air navigation service providers [6]. These converging trends provide the essential basis for the development of large-scale, ubiquitous, trusted, controllable, and operational LAENets.

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Multiple research efforts have explored over LAE-Nets. Telecom operators have proposed integrated sensing and communication (ISAC) architectures based on 5G-A; aerospace and geospatial communities focus on fusing radar for high-precision positioning; and air navigation service providers promote predefined airspace grids and structured routing. While effective in specific contexts, these efforts often operate in silos, targeting either sensing, communication, or routing, without a unified, holistic framework that integrates sensing, connectivity, localization, management and service. This fragmentation limits their applicability in complex, large-scale UAV operations and airspace management. Recent safety statistics corroborate this view: The European Aviation Safety Agency (EASA) Annual Safety Review Report indicates that EU member states recorded 12 drone system accidents caused by system failures in 2023; the UK Air Accidents Investigation Branch's 2024 Annual Safety Review Report categorized 54 drone incidents; while the Federal Aviation Administration (FAA)'s 2023 Unmanned Traffic Management (UTM) field tests uncovered numerous instances of non-standard operating practices, highlighting how siloed designs struggle to meet large-scale reliability and interoperability requirements.

To fill this gap, we propose a Ubiquitous and Trusted Intelligent Cellular-native Network (UTICN) for the LAE. UTICN adopts a unified cellular-native architecture that integrates sensing, positioning, communication, management, and service to provide support for safe, ubiquitous, trusted, controllable, and operational LAE. It features key enabling technologies such as ISAC, active and passive positioning, air-ground intelligent machine communication, control-data decoupled management, and swarm intelligence-driven situational evolution. Our proposed solution is validated through two representative deployments: a city-scale LAE management system and a cooperative multi-frequency ISAC system.

The remainder of this paper is organized as follows. Section II outlines the key requirements for UTICN. Sections III and IV introduce the UTICN architecture and elaborate on its core components and enabling technologies. Section V presents two typical application cases. Finally, Section VII concludes this work.

II. LAE REQUIREMENTS

LAE demands the development of a ubiquitous and trusted infrastructure. This infrastructure should facilitate interconnection among UAVs and relevant stakeholders, such as government authorities, UAV service providers, and air navigation service providers. It must also ensure global observability and governance, persistent manageability, and timely controllability of low-altitude operations.

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Despite increasing UAV activity, LAE systems still suffer from foundational challenges summarized as invisible, disconnected, and uncontrollable. These

issues stem from the absence of a unified and holistic framework that spans the full lifecycle, operational chain, and stakeholders. Currently, UAVs predominantly rely on isolated point-to-point communication links, executing pre-planned missions with limited real-time environmental awareness and without dynamic access to airspace constraint information. As a result, most systems cannot reliably identify abnormal targets, trace their sources, or respond effectively to unauthorized behaviors, accumulating substantial operational risks.

To enable safe and orderly UAV operations and airspace management across diverse scenarios, it is essential to establish a unified and holistic platform that provides dynamic airspace management, unmanned aerial vehicle operation, and safety management. The platform must support the full process, from mission approval and identity registration to task execution, while ensuring seamless cross-domain networking, dynamic link handover, and system interoperability. Moreover, it must enable timely responses and effective control of abnormal and unsafe events, such as unauthorized UAV activities (e.g., rogue drones) [8]. Meeting these goals requires that the platform satisfy three core capabilities:

Sensing requirements: multi-dimensional real-time awareness spanning both the physical and the electromagnetic space. In the physical space, the sensing system must deliver continuous, wide-area, and fine-grained coverage, capable of operating in complex urban settings with dense buildings and dynamic weather conditions. It should support all-weather, all-domain monitoring, enabling accurate identification, classification, and tracking of both legitimate and rogue UAVs. Furthermore, region-level situational awareness is essential for monitoring high-risk or sensitive areas, capturing traffic density, trajectory deviations, and emergent threats to support large-scale, coordinated UAV scheduling. In the electromagnetic space, the sensing infrastructure must provide real-time visibility into the spectral landscape of the low-altitude airspace. This includes monitoring authorized communication links, identifying abnormal or malicious signals, and analyzing electromagnetic behaviors across frequency, time, and spatial dimensions. Robust electromagnetic sensing is critical to ensure reliable communications, detect unauthorized transmissions, and support timely safety interventions and public security enforcement.

Communication requirements: high reachability, stability, and mission-critical connectivity. With the increasing number and intelligence level of UAVs, the communication system must go beyond basic support for UAV control and task data transmission. It should enable efficient state exchange and coordination among UAVs, as well as among UAVs, LAE infrastructures, and relevant stakeholders. Especially in complex urban environments, communication links must maintain high accessibility and robustness, ensuring UAVs can continuously receive regulatory commands, upload operational states, and promptly respond to airspace scheduling and emergencies. Moreover, the system should support diverse communication modalities such as air-to-air and air-to-ground links, enabling unified access and hierarchical management across different tiers and types of airspace participants. It must also incorporate differentiated protection mechanisms for control signaling and mission payloads, ensuring higher priority and reliability for regulatory instructions.

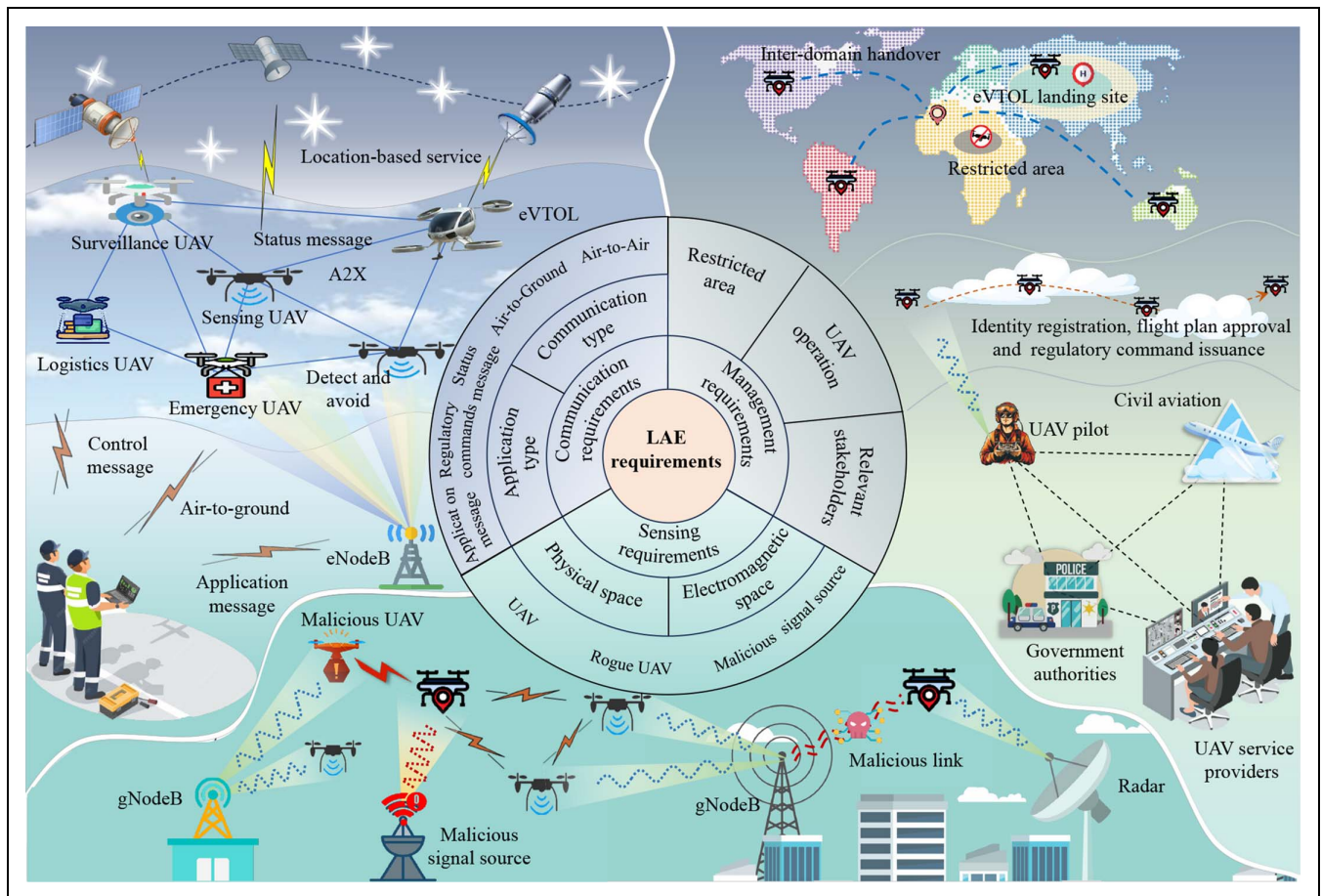


FIG. 1. LAE requirements, including sensing, communication, and management requirements.

Management requirements: LAENets should support full-process management from identity registration and flight plan approval to real-time airspace monitoring and command issuance. It must enable rapid identification and intervention of abnormal behaviors and allow coordinated regulation across government authorities, UAV pilots, and air navigation service providers. To build a sustainable and scalable low-altitude regulatory framework, the system must fulfill three essential capabilities: airspace observability to monitor flight environments, operational status, and abnormal targets; operational manageability to support approval, scheduling, alerts, and trajectory guidance; and behavioral controllability to deliver commands and enforce interventions against illegal or unsafe operations.

We propose a Cellular-Native Integrated Architecture for Low-Altitude Economy (UTICN) designed to overcome the bottlenecks of existing systems operating in silos and lacking a unified framework: (1) At the system level, UTICN integrates perception, positioning, communication, management, and services onto a single foundation. It employs a design decoupling the control plane from the data plane, forming a closed-loop system capable of global observation, management, and intervention; (2) At the key technology level, it integrates multi-source, multi-modal sensing with electromagnetic situation monitoring. Combining passive and active positioning methods (e.g., GNSS with cellular carrier phase, radar with vision) achieves high precision and reliability in complex urban environments. Simultaneously, it provides continuous, robust mission-critical connectivity and

efficient data transmission through unified air-to-air and air-to-ground access, heterogeneous technology coordination (RedCap and direct connectivity), and mission-differentiated assurance. (3) At the operational level, establish a layered, domain-specific service orchestration and coordination mechanism to support cross-regional, cross-entity approval, scheduling, alerting, and risk management. Validate its feasibility and scalability through city-level low-altitude airspace management systems and multi-frequency collaborative ISAC prototypes, laying the digital foundation for a large-scale, trustworthy, controllable, and operational LAENet.

III. UTICN ARCHITECTURE

In this section, we will provide a detailed introduction to the core functions of UTICN. As illustrated in Fig. 2, UTICN integrates five core functional subsystems, i.e., sensing, positioning, communication, management, and service, into a unified, biquitous, trusted, controllable, and operational infrastructure for next-generation airspace management and UAV operations.

Sensing subsystem: Enables multi-domain situational awareness by integrating multi-modal sensing technologies over both the physical space and the electromagnetic space, including radar, infrared vision, electromagnetic sensing, A2X communication, and ISAC. ISAC, a key enabler in 6G systems, utilizes embedded reference signals to support real-time sensing over UAVs, airspace, and infrastructures. In addition, electromagnetic environment sensing enables interference localization and signal integrity

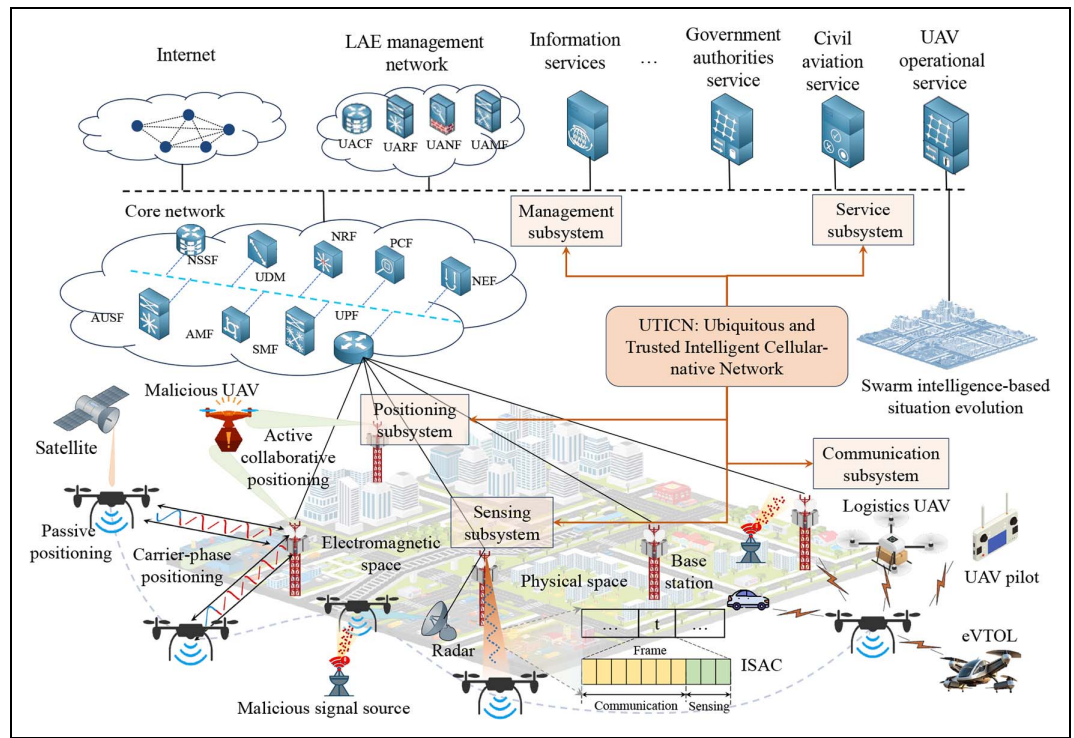


FIG. 2. UTICN, a ubiquitous and trusted intelligent cellular-native network, which integrates five core functional subsystems, i.e., sensing, positioning, communication, management, and service subsystem. It provides a systematic solution for UAV regulation, UAV operational services, airspace management, and public security management.

monitoring, ensuring robust support for both navigation and communications.

Positioning subsystem: Provides high-precision and continuous localization by integrating passive, e.g., Global Navigation Satellite System (GNSS) and real-time kinematic differential positioning, and active, e.g., cellular carrier-phase methods, and radar/vision-based techniques, positioning technologies. Cellular carrier-phase techniques deliver centimeter-level accuracy in GNSS-challenged environments (e.g., tunnels or dense urban areas). Through multi-source fusion, the system refines coarse estimates and enables accurate, reliable positioning essential for UAV operations and control in complex scenarios.

Communication subsystem: Supports wide-area and mission-critical air-ground and air-air communications, which are critical for UAV operation, airspace management, data transmission, and cooperative mission execution. It integrates dedicated data links, 5G RedCap communication, PC5-based A2X direct connectivity [9]. On transmitting large-volume data, semantic communication selectively transmits mission-relevant data, thereby improving spectral efficiency under constrained conditions.

Management subsystem: Establishes an independent and resilient UAV control plane that is decoupled from mission-oriented data flows. Functional modules such as UAV Access Gateway Function (UAGF), UAV Access Control Function (UACF), UAV Registration Function (UARF), UAV Application Management Function (UAMF), and UAV Application Network Function (UANF) provide full lifecycle management for UAVs. This layer supports multi-standard access, cross-domain identity coordination, and rapid incident response, enabling regulatory oversight, service authentication, and control command dissemination in real time.

Service subsystem: Provides an orchestration layer for multi-role operations, integrating UAV task execution, supervisory control, and public safety services. With hierarchical deployment strategies spanning regional, provincial, and national levels, it supports customized, scenario-specific services while ensuring interoperability across UAV service domains. This enables dynamic route planning, adaptive task scheduling, and rapid risk mitigation across diverse LAE applications.

Subsystem coordination: The sensing subsystem performs multi-modal probing of physical and electromagnetic space and produces timestamped detections and signal descriptors. The positioning subsystem fuses GNSS or RTK, cellular carrier-phase, and radar or vision to generate state estimates and uncertainty metrics, which are time-aligned with sensing outputs. The communication subsystem transports these streams to the edge and core with QoS isolation for command and control, using Uu for air-ground and PC5 for air-air exchange; it also delivers assistance data to positioning and tasking messages back to sensing. The management subsystem authenticates identities, validates flight intents, maintains an airspace graph or digital twin, executes deconfliction and policy checks, and issues control actions such as reroute, hold, return-to-home, or landing. The service subsystem orchestrates missions and resources across domains, exposes APIs to stakeholders, and requests targeted sensing, bandwidth boosts, or higher update rates when mission risk rises. Feedback is continuous: management and service retask sensing, communication adapts scheduling and handover, positioning refines estimates with network assistance, and detected anomalies trigger management alarms and service-level mitigation.

Interworking with terrestrial mobile networks: UTICN is a logically independent yet 3GPP-compliant

management network that interworks with public 5G systems via UAGF and NEF/PCF policy exposure. Service continuity is ensured by make-before-break hand-over with PSA-UPF anchors, slice-aware inter-PLMN roaming with policy transfer and UPF relocation, and multi-access steering with failover between PC5 and Uu for prioritized C2. Traffic scheduling separates control and payload by slicing and QoS flows: C2 uses GBR with pre-emption and configured-grant RAN scheduling, while payload is steered by UPF to MEC or cloud according to policy and congestion. A zero-trust control plane with authentication, encryption, and auditable overrides maintains supervisory authority during inter-domain operations.

In summary, UTICN delivers a vertically integrated architecture that closes the loop from low-altitude situational sensing to decision-making and real-time control. Its cellular-native design forms a secure and intelligent digital foundation to ensure the safety, scalability, and serviceability of future low-altitude economic systems.

IV. KEY ENABLING TECHNOLOGIES

To realize UTICN, it is essential to overcome challenges across five core domains: sensing, positioning, communication, management, and service. This section introduces the key enabling technologies for UTICN.

A. MULTI-DOMAIN INTEGRATED SENSING

Comprehensive sensing of low-altitude airspace requires coverage of both the physical space and the electromagnetic environment. As for the physical space, UTICN supports ISAC, which reuses spectrum and RF chains to support simultaneous communication and radar sensing. Using in-band coupling, radar symbols are embedded within control resource blocks. This design enables real-time tracking of small UAVs while maintaining high-throughput data links [10], [11]. Our field tests demonstrate 15 ms air-interface latency, 2 Gbps throughput, and 1.5 m range resolution, with successful multi-band cooperative detection of UAVs with radar cross-sections as low as 0.1 m^2 .

Low-altitude airspace is also characterized by dense electromagnetic activity, where communication, navigation, and sensing signals often interfere with one another, especially in urban environments. To ensure robust situational awareness, UTICN integrates electromagnetic sensing across radar, optical, and radio frequency (RF) domains for seamless target tracking and interference localization.

To further enhance electromagnetic awareness, UTICN employs multi-domain analysis methods. These include automatic modulation classification to recognize UAV control links, radiation source fingerprinting across time-frequency-space domains, and blind protocol parsing to decode unknown signal formats. UTICN couples RF fingerprinting, automatic modulation classification, and blind protocol parsing with interference-source localization to detect rogue emitters and preserve link integrity under dense urban EMI. Finally, UTICN builds a unified multi-domain representation model using deep neural networks to capture emitter characteristics across multiple dimensions. Furthermore, dynamic control information is extracted to infer user behaviors and identify anomalies.

Compared with radar-only or optical-only pipelines and standalone spectrum monitors, UTICN provides a unified multi-domain sensing stack. It leverages integrated sensing and communications to reuse spectrum and RF chains for joint communication and sensing, aligning with ongoing 3GPP studies that formalize ISAC channel models and deployment scenarios. On the electromagnetic side, UTICN layers automatic modulation classification, RF fingerprinting, and blind protocol analysis to recognize control links and detect rogue emitters, which surpasses single-feature detectors in cluttered urban environments. It further incorporates interference-source localization using timing-based and ray-tracing-aided methods suited to dense cities, enabling faster attribution and mitigation while preserving link quality through in-band coupling.

B. MULTI-SOURCE COLLABORATIVE POSITIONING

Accurate positioning in urban low-altitude environments is challenged by satellite signal occlusion from buildings, tunnels, and underground infrastructure. To ensure robust UAV tracking, UTICN adopts a multi-source fusion approach combining GNSS, ground-based differential corrections, pseudolite systems, and mobile network-based positioning, providing both active and passive positioning.

With recent breakthroughs in millimeter-wave communications and massive multiple-input multiple-output (MIMO), cellular carrier-phase positioning significantly improves ranging and angular resolution, offering centimeter-level positioning accuracy [12]. Carrier-phase positioning extracts phase information from orthogonal frequency-division multiplexing (OFDM) multi-carrier signals. By resolving integer ambiguity and correlating phase with signal wavelength, UTICN achieves sub-decimeter accuracy. Using differential techniques across multiple ground stations, measurement noise is mitigated, providing higher positioning accuracy.

Depending on deployment, positioning can be calculated either actively, by aggregating measurements from ground base stations, or passively, with UAVs performing onboard localization using GNSS or real-time kinematic differential positioning signals. With improvements in onboard chips, carrier-phase positioning is feasible and scalable for UAV active positioning. To further enhance spatial resolution, UTICN implements multi-station active collaborative localization. ISAC-enabled base stations capture both self-transmitted echoes and echoes from neighboring stations. By leveraging inherent time-delay correlations, both active and passive signals are fused to reinforce positioning precision. UTICN hardens localization against spoofing and jamming via cross-domain consistency checks between GNSS, cellular carrier-phase, pseudolites, and ISAC echoes, with integrity flags for downstream control.

This approach yields three gains. First, coherent signal stacking significantly improves the signal to noise ratio. Second, passive signals synthesized from active measurements suppress noise and enhance robustness. Third, sequential extension with phase linked interpolation improves spatial resolution. In contrast, methods that rely only on GNSS or RTK in urban canyons often exhibit large non line of sight and multipath biases, frequently on the order of tens of meters and in documented cases exceeding 50 meters in dense street canyons, which disrupts continuity for low altitude tracking; UTICN mitigates these

errors through cross station measurements. UWB only localization also degrades under non line of sight, with meter level ranging biases and a sharp drop in the probability of achieving submeter error, which limits reliability in cluttered corridors. Single station 5G or millimeter wave positioning can reach centimeter level accuracy under favorable line of sight, but real world measurements reveal coverage holes and blockage sensitivity that require dense deployments to sustain continuity; UTICN's multi station integration counteracts such outages. By fusing active and passive measurements across multiple stations, UTICN reduces non line of sight induced bias and tightens error bounds relative to single anchor time of arrival baselines in mixed line of sight and non line of sight settings, enabling accurate and continuous UAV tracking in complex urban environments.

C. INTELLIGENT MACHINE COMMUNICATION

As the low-altitude economy scales, conventional UAVs relying on direct links over unlicensed bands (e.g., 2.4 GHz and 5.8 GHz) face increasing spectrum congestion and interference, particularly in urban environments. While proprietary data links remain useful, the proliferation of UAVs necessitates scalable, interference-resilient, and mission-critical connectivity solutions. UTICN addresses this by leveraging the cellular network infrastructure, dynamically adjusting antenna beam patterns to optimize coverage in near-ground airspace. The integration of 5G-A and ISAC technologies further strengthens air-ground connectivity, enabling wide-area, high-reliability links with situational awareness.

To balance performance and efficiency, UTICN employs 5G NR RedCap for mid-rate transmission tasks. RedCap, standardized in 3GPP Releases 17 and Releases 18, reduces terminal complexity by constraining bandwidth, modulation schemes, and antenna count, resulting in up to 60% cost and power savings while preserving key 5G capabilities such as low latency and edge intelligence. RedCap's compatibility with network slicing makes it ideal for UAV control and telemetry applications. Control UAV is logically isolated from mission payloads through network slicing and UPF-based routing, ensuring secure and differentiated service delivery.

Beyond air-ground links, UTICN introduces A2X technology to enable direct UAV-to-UAV communication, addressing challenges such as collision avoidance and swarm coordination. Built upon the evolution of Cellular Vehicle-to-Everything, A2X employs PC5 sidelink interfaces standardized in 3GPP Release 19 [13]. It supports unicast, multicast, and broadcast over the 5.9 GHz, with NR-V2X protocol stack compatibility. A2X achieves sub-3 ms latency and over 1 km communication range in open environments, significantly outperforming legacy protocols such as Wi-Fi mesh or LoRa, which suffer from interference and limited throughput.

A2X also provides a unified communication framework for both air-to-air and air-to-ground interactions. UAVs can share identity, trajectory, and direct detect-and-avoid information with relevant stakeholders. This enables fine-grained trajectory deconfliction, formation control, and real-time situational awareness. Through integration with Uu interfaces and edge A2X service nodes, UAV controllers can maintain command-and-control links across network domains, ensuring continuous supervision and

responsive control even in beyond-visual-line-of-sight scenarios.

By incorporating A2X into UTICN, UAVs gain not only autonomous negotiation capabilities but also network-assisted coordination via cellular base stations. This supports scalable UAV swarming, distributed perception, and multi-agent collaborative decision-making [14]. Furthermore, A2X lays the foundation for a parallel, regulation-aware control network decoupled from mission data flows, enhancing resilience and enabling cooperative autonomy in dense, contested low-altitude environments.

For high-fidelity video transmission under bandwidth constraints, UTICN supports semantic communication techniques. Unlike traditional source-channel separation, semantic communication extracts and transmits task-relevant information based on shared knowledge between sender and receiver, significantly reducing data volume. By deploying pretrained semantic encoders/decoders, UAVs can achieve up to 60% data reduction in real-time video streaming without perceptual degradation, thus enhancing communication robustness in contested low-altitude environments. With advances in lightweight AI models and onboard acceleration, semantic communication emerges as a key enabler for future bandwidth-efficient UAV operations.

In contrast to proprietary links over unlicensed 2.4 GHz and 5.8 GHz, which have no interference protection and must accept interference, UTICN carries control and payload on licensed cellular spectrum with schedulable QoS and network slicing, enabling interference-resilient, mission-critical delivery. In air-to-air coordination, NR V2X over PC5 achieves lower latency and a longer reliable range than IEEE 802.11p in reported evaluations, and UTICN integrates this sidelink with Uu-based supervision to maintain continuity across domains. Slice-level isolation, authenticated NR V2X sidelink, and multi-link failover between PC5 and Uu provide prioritized, jam-resilient C2 with rapid fallback paths. For mid-rate telemetry, NR RedCap reduces device complexity and power while preserving key 5G capabilities and slice compatibility, which supports cost-efficient, fleet-scale command and control and telemetry. With 5G-Advanced, carrier-phase positioning becomes part of the standardized toolkit; when fused with ISAC measurements in UTICN, it provides centimeter-class accuracy under favorable conditions and improves continuity in GNSS-challenged urban airspace.

D. INTELLIGENT CONCISE MANAGEMENT

To enable secure, scalable, and mission-critical UAV operations, UTICN introduces a dedicated control-data separation framework that decouples the UAV control plane from mission data flows. This architecture is inspired by the Aeronautical Telecommunication Network used in civil aviation, where UAVs operate outside traditional air traffic control systems and coexist with terrestrial public networks.

A specialized low-altitude UAV management network is constructed independently from the public internet, ensuring high-reliability services such as trajectory approval, cross-domain coordination, flight status monitoring, conflict avoidance, and emergency intervention. Functional modules, e.g., UAGF, UACF, UARF, UAMF, UANF, collaboratively support seamless UAV registration, identity synchronization, mobility management, and inter-domain handover. For

example, UAGF provides a unified interface adaptation and seamless access across domains.

The architecture supports heterogeneous communication modalities, including 5G-A/A2X cellular interfaces, PC5-based direct communication, and proprietary UAV links. To address regulatory constraints, the network supports localized control through hierarchical service domains, i.e., district, provincial, and national, ensuring governance compatibility with public safety, civil aviation, and military authorities.

To meet the real-time demands of UAV operations, the control plane integrates low-latency relay via edge UAGF, avoiding cloud backhaul and reducing control response times. It also supports audit trails and override commands such as forced hovering, return-to-home, or landing in case of emergencies, ensuring enforceable regulatory supervision. A zero-trust control plane with mutual authentication, end-to-end encryption, edge anchoring, audit trails, and policy-driven overrides enables enforceable, low-latency supervisory control.

Additionally, UTICN integrates mobile edge computing (MEC), UPF, network slicing, and QoS-aware orchestration to enable differentiated traffic routing for UAV control and mission payloads. UAVs dynamically access control and service providers, while UPF performs content-based steering toward MEC or clouds based on latency and bandwidth requirements. This architecture guarantees secure isolation and scalable service delivery across distributed UAV deployments.

As UAV operations evolve toward swarm-based collaboration and AI-powered autonomy, UTICN supports hierarchical offloading strategies. Real-time control is kept on board, mission-critical computation is offloaded to MEC, and bulk tasks such as environmental modeling or multi-agent learning are routed to the cloud. A unified view of spectrum availability, computational load, and spatial-temporal airspace dynamics is maintained to support coordinated planning across physical, communication, and computing dimensions.

Unlike the Aeronautical Telecommunication Network (ATN), which is specified for air traffic services communications in manned aviation and implemented with technologies such as VDL Mode 2 and the Aeronautical Mobile Airport Communication System (AeroMACS) on the airport surface, low-altitude drone operations are managed through UTM/U-space digital services outside conventional Air Traffic Control (ATC) separation. Consequently, direct integration of low-altitude UAVs into ATN is not aligned with current operational or regulatory models. UTICN addresses this gap with a cellular-native control and data separation design anchored by MEC, UPF, and network slicing, providing deterministic QoS, low latency with termination at the edge, and governance-aligned hierarchical domains for public safety and civil aviation oversight. It further unifies Uu and PC5 NR V2X to deliver continuous air-to-air and air-to-ground supervision across domains, a level of continuity that shared best effort networks cannot guarantee at scale.

E. SWARM INTELLIGENCE-BASED SITUATION EVOLUTION

The rapid increase in UAV density renders static, pre-planned routing strategies ineffective in managing large-scale LAE UAVs. Traditional traffic prediction methods are limited by the lack of historical data,

while centralized multi-agent trajectory planning suffers from exponential complexity due to inter-UAV interactions in 3D airspace.

To address this, inspired by the traffic situation evolution in terrestrial transportation [15], UTICN introduces a hierarchical situation evolution framework that integrates evolutionary game theory and Monte Carlo Tree Search (MCTS). Locally, UAVs use game-theoretic strategies to achieve user equilibrium, where each agent selects its optimal path based on local congestion and delay feedback. Globally, MCTS enables efficient exploration of multi-agent path combinations, leveraging pruning and risk evaluation to rapidly derive near-optimal swarm trajectories.

Compared with data-driven traffic prediction that requires extensive historical corpora, centralized multi-agent planners with exponential search, and large-scale reinforcement learning with prohibitive training costs in 3D airspace, the UTICN framework operates without heavy data dependence and with tractable complexity. By decomposing global coordination into local game-theoretic equilibria and a global MCTS exploration with pruning and risk evaluation, UTICN reduces the search burden while retaining near-optimality under operational time budgets. Execution at the network edge and state dissemination over A2X and Uu enable minute-level adaptation, regulatory constraint enforcement, and compatibility with heterogeneous onboard autonomy. Signed intent messages, rate-limited actuation, and Byzantine-robust consensus guard against malicious or faulty agents during coordinated trajectory updates. As a result, UTICN provides scalable and robust trajectory coordination for dense low-altitude operations.

V. APPLICATION CASES

A. CASE 1: CITY-LEVEL LAE MANAGEMENT PLATFORM

A practical implementation of UTICN's management function is shown in Fig. 3. This system presents a pilot deployment in Guangzhou, China, serving as a benchmark for future low-altitude platforms. This system is designed for airspace flight governance and establishes a closed loop that comprises airspace sensing, state cognition, risk identification, and supervisory override of UAVs. The sensing layer leverages the networked detection capability enabled by integrated sensing and communications in Case 2 together with cooperative A2X communications between air and ground. The subsequent stages of cognition, risk assessment, and override are realized through the air to ground communications network combined with programmable control over UAVs. Importantly, the architecture separates supervisory override at the regulatory level from mission data transport at the system level, a separation that has not been fully achieved in existing systems.

Airspace and UAV operations are administered through fine-grained delineation of control domains and classification of flight behaviors, supported by diverse routing modes (fixed, dynamic, or free) and tiered monitoring from localized pads to city-scale vertiports. When onboard intelligence allows the air traffic management system to orchestrate missions, UAVs follow low-altitude governance guidance for takeoff, landing, and trajectory planning, which safeguards airspace safety and maximizes utilization. When UAVs lack such onboard autonomy and remain remotely piloted via ground systems, the

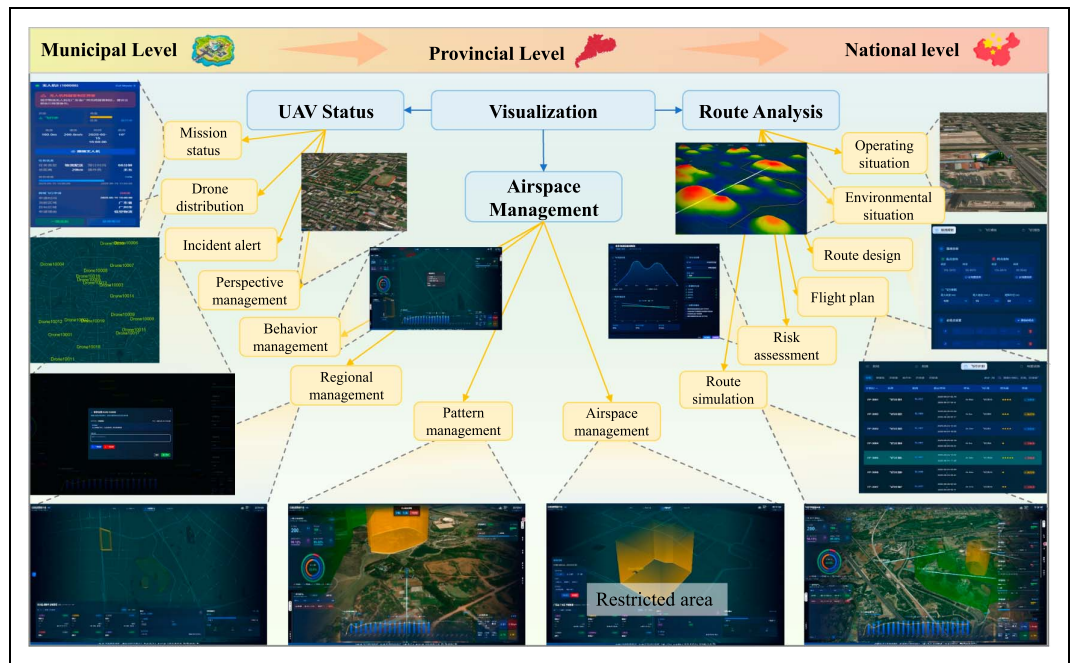


FIG. 3. Visualization of the function for the city-level LAE management platform.

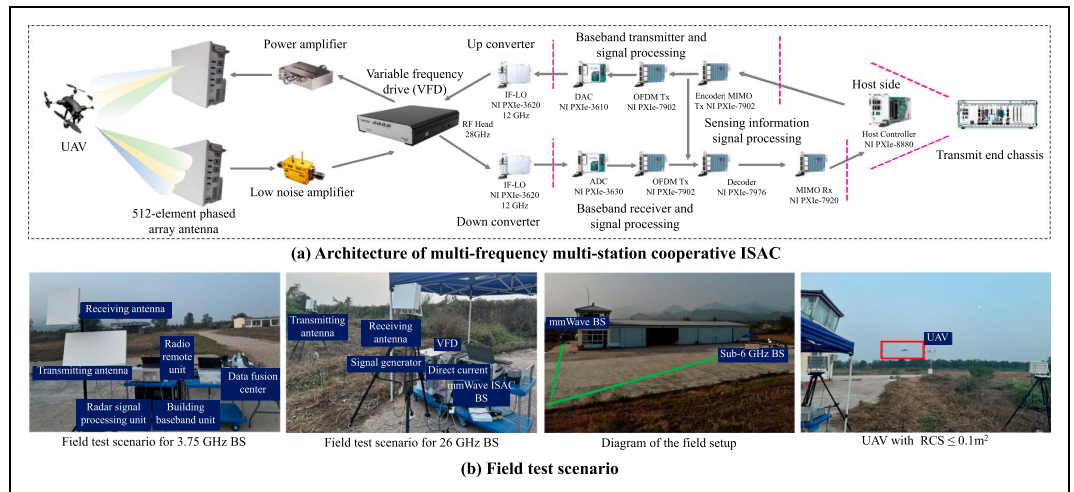


FIG. 4. Design and field test of the prototype multi-frequency multi-station ISAC system.

low-altitude control network will provide supervisory override so that, under airspace restrictions or anomalies, it can issue commands to hold in place, return to home, or land, thereby ensuring effective risk control.

The system provides a multi-level visualization interface (i.e., national, provincial, district), offering comprehensive insight into UAV distribution, mission status, alert conditions, and regional traffic dynamics. Users can interactively inspect and manage UAVs, flight paths, airspace conditions, and event responses.

The route analysis engine supports both automatic and manual planning, enabling situational evolution to address low-altitude airspace awareness challenges. It facilitates swarm-based intelligent flight path planning for multiple UAVs, enhancing low-altitude flight safety.

The platform also features regional resource management, enabling visualization and dynamic adjustment of airspace, ground infrastructure, and operational boundaries. New zones can be added or adjusted with automatic alerts on their impact on existing flight routes.

Finally, the system integrates meteorological and geographic situational analysis, visualizing the effects of rainfall, wind, terrain, and signal quality. We adopt 3D grid models to assess flight suitability, offering situational support for the coordinated and safe operation of large UAV fleets.

B. CASE 2: MULTI-FREQUENCY COLLABORATIVE ISAC

To validate UTICN's sensing and communication effectiveness in complex urban environments, we developed a prototype platform based on a multi-frequency ISAC-enabled collaborative sensing system. The system architecture (Fig. 4(a)) includes a baseband processor that maps user data into ISAC baseband signals, which are then upconverted to millimeter-wave or Sub-6 GHz frequencies and transmitted via high-power antennas. These waveforms simultaneously support radar-based object sensing and wireless communication.

Two ISAC base stations (BS) were deployed in our field test, i.e., Sub-6 GHz BS for wide-area sensing, working at 3.75 GHz, with 100 MHz bandwidth and

46 dBm effective isotropic radiated power (EIRP), and millimeter-wave (mmWave) BS for close-range high-resolution detection, working at 26 GHz, with 800 MHz bandwidth and 55 dBm EIRP. Both systems used OFDM waveforms with subcarrier-embedded radar samples for sensing-communication integration. The mmWave BS also employed massive MIMO and digital self-interference suppression to ensure precise signal reception. The sensing results of these two BSs are fused to form the observed trajectory of the target UAV. More detailed parameter settings are shown in Table I.

In the experiment (Fig. 4(b)), a UAV with radar cross-sections less than 0.1 m^2 flew north to south, initially detected only by the Sub-6 GHz BS. Upon entering an overlapping coverage zone, both stations contributed to cooperative sensing. As the UAV moved eastward, the mmWave BS became the primary detector due to urban obstructions affecting the sub-6 GHz BS. A centralized processor fused multi-station data to generate a unified UAV trajectory.

System performance was evaluated using ground-truth GPS trajectories. Positions with under-10-meter error were considered effective, following 3GPP TR22.837 standards. All 80 sampled trajectory points met this criterion, as shown in Fig. 5(a). As illustrated in Fig. 5(b), Sub-6 GHz BS showed a mean error of 2.5 meters, the mmWave BS 3.5 meters, and the fused estimate just 2.0 meters. Furthermore, this experiment demonstrates that fusion over the mmWave BS and Sub-6 GHz BS enhanced continuity and tracking robustness under partial obstruction.

This case study confirms the feasibility and superior accuracy of multi-frequency, ISAC-enabled cooperative sensing in dense urban environments. The system offers valuable insights for applications in low-altitude traffic management, urban security, military reconnaissance, and disaster response.

VI. FUTURE RESEARCH DIRECTIONS

Despite the comprehensive capabilities of UTICN, several technical limitations and open challenges remain. The UTICN architecture was designed to provide a unified framework for low-altitude UAV networking, yet fully realizing its vision will require addressing issues related to cost, scalability, spectrum management, real-time coordination, standardization, and security. The authors of UTICN themselves note that further research is needed in areas such as dynamic topology optimization, adaptive spectrum management, and multi-tier resource coordination to achieve a seamless, resilient low-altitude network infrastructure. In this section, we objectively analyze these limitations and risks, and outline future research directions to overcome them in the evolution of UTICN.

Infrastructure Deployment Challenges in Urban and Peri-Urban Areas: A major practical concern for UTICN is the cost and complexity of infrastructure deployment across heterogeneous environments. Achieving ubiquitous low-altitude coverage and sensing in dense cities may require additional ground assets, such as denser base-station grids, radars, and landing hubs, which impose significant upfront investment. Despite advances in cellular networks, urban low-altitude links still suffer building-induced blockage and interference, so a robust and scalable infrastructure is needed to sustain reliable connectivity. Peri-urban and rural areas often lack adequate

Module	Parameter	Value
mmWave BS	Band	26 GHz
	Start/Stop frequency	25.5–26.5 GHz
	System bandwidth	800 MHz
	Antenna EIRP	55 dBm
	Symbol PRI	0.2 ms
	Subcarrier spacing	75 kHz
Sub-6 GHz BS	Band	3.75 GHz
	Start/Stop frequency	3.7–3.8 GHz
	System bandwidth	100 MHz
	Antenna EIRP	46 dBm
	Symbol PRI	2.5 ms
	Subcarrier spacing	30 kHz
	UAV radar cross-sections	$\leq 0.1 \text{ m}^2$
Data fusion center refresh rate		1.5 s

TABLE I. Field test parameter settings.

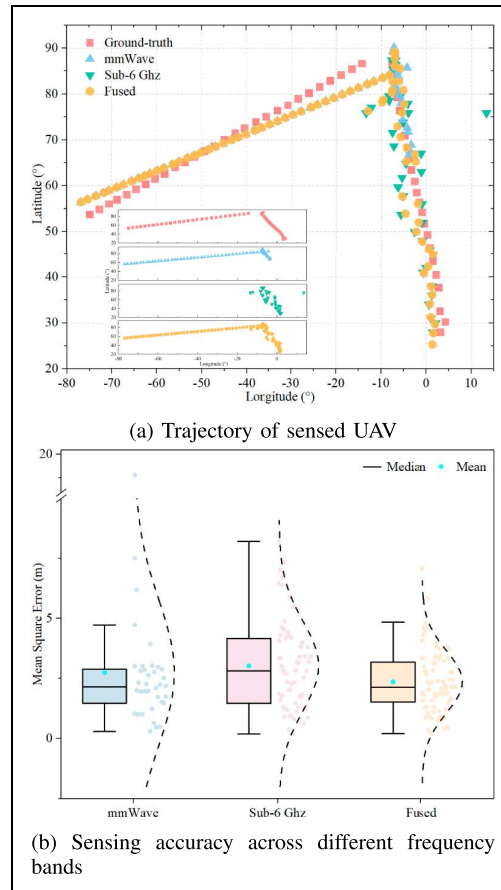


FIG. 5. Performance of multi-frequency collaborative ISAC.

communications, creating coverage gaps. A promising direction is a multi-tier architecture that integrates terrestrial and non-terrestrial assets. For example, low Earth orbit constellations can complement ground networks by providing wide-area, low-latency links where infrastructure is sparse. By orchestrating air-ground and satellite communications, UTICN can extend coverage without extensive ground build-out. Future work should optimize infrastructure layouts and cost models, such as joint

base-station siting and resource placement tailored to low-altitude traffic patterns, to minimize cost while ensuring capacity across both urban and sparsely populated airspace.

Scalability and Energy Efficiency for Large-Scale UAV Operations: As UTICN scales to support large fleets of UAVs and eVTOL aircraft, scalability and energy sustainability become critical. High vehicle counts will increase traffic loads and require frequent recharging or refueling, stressing both network and power infrastructures. To mitigate these pressures, future work should pursue energy-aware network design and aerial energy delivery. One direction is aerial charging and relay networks in which solar powered charging UAVs or high-altitude platforms act as mobile stations to form an air-to-air energy supply chain. Complementary approaches include advanced energy harvesting (for example, biohybrid fuel cells) and wireless power transfer using millimeter wave or laser links to extend endurance. On the network side, scalable coordination algorithms are needed so that control systems remain stable as fleet size grows. Generative AI based deep reinforcement learning can jointly optimize task scheduling and power allocation in real time according to mission needs and battery states. In combination, improved energy sources, in situ recharging, and intelligent power management can enable UTICN to scale economically to citywide and nationwide operations without compromising reliability.

Spectrum Availability and Interference Management in Low-Altitude Environments: The low-altitude airspace presents distinct spectrum challenges for UTICN. Unlike traditional terrestrial networks, UAV links at low altitude face dynamic electromagnetic conditions, including interference from ground systems, building-induced multipath, and emissions from other aerial platforms. Spectrum is scarce, and uncoordinated use can degrade control links and throughput. UTICN therefore requires real-time spectral awareness with continuous monitoring of authorized links and detection of anomalous or malicious signals across frequency, time, and space.

Future work should develop AI-enabled spectrum management that adapts to interference. Promising directions include machine-learning models for spectrum prediction and interference cartography, followed by cognitive-radio reconfiguration and channel hopping to avoid congestion and jamming. Cooperative architectures that integrate interference modeling with AI-driven decision making can improve reliability. Complementary advances are needed in interference-aware routing that dynamically selects frequencies and paths, together with the joint use of dynamic spectrum access, interference cancellation, and multi-link routing. Rigorous evaluation under realistic low-altitude scenarios is essential to ensure that safety-critical control traffic remains uninterrupted.

Data Security and Privacy in Control-Data Separation Architectures: The low-altitude airspace presents distinct spectrum challenges for UTICN. Unlike traditional terrestrial networks, UAV links at low altitude face dynamic electromagnetic conditions, including interference from ground systems, building-induced multipath, and emissions from other aerial platforms. Spectrum is scarce, and uncoordinated use can degrade control links and throughput. UTICN therefore requires real-time spectral awareness with continuous monitoring of authorized links

and detection of anomalous or malicious signals across frequency, time, and space.

Future work should specify a multilayer security framework for UTICN. Coordinated controls across cloud, edge, and device with trusted execution environments, edge intrusion detection, and end to end encryption can protect commands and data even if one segment is compromised. Privacy and resilience should rely on differential privacy, federated learning, autonomous anomaly detection, rapid reconfiguration, and path diversity to maintain confidentiality, integrity, and availability at scale.

VII. CONCLUSION

This article presents UTICN, a ubiquitous and Trusted Intelligent Cellular-native Network architecture purpose-built to support the intelligent, large-scale, ubiquitous, trusted, controllable, and operational development of LAE. By integrating multi-domain sensing, high-precision multi-source positioning, intelligent air-ground communication, control-data separated management, and swarm intelligence-driven situation evolution, UTICN offers a unified framework to address the pressing challenges of dynamic airspace management, UAV operation, and security management in complex low-altitude environments.

Through a modular cellular-native architecture and cross-domain technological integration, UTICN lays a robust foundation for future applications in urban airspace management, mission-critical UAV coordination, public security operations, and disaster response. As LAE systems continue to evolve, further research is needed in dynamic topology optimization, adaptive spectrum management, and multi-tier resource coordination. These efforts will enable seamless integration between terrestrial, aerial, and space-based networks, paving the way toward a fully connected, resilient, and context-aware low-altitude infrastructure ecosystem.

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